

Supply Chain & Logistics Analytics Report

Project Overview

The objective of this project was to analyze supply chain operations and identify factors affecting delivery performance and profitability. Using data analytics and machine learning techniques, the project provides actionable insights to improve operational efficiency and reduce delivery delays.

Business Problem

Supply chain organizations often face challenges such as:

- Late deliveries
- Increased logistics costs
- Reduced customer satisfaction
- Operational inefficiencies
- Profitability losses

This project aims to identify the root causes of these issues and support data-driven decision-making.

Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Power BI

Data Preparation

The dataset was cleaned and prepared through:

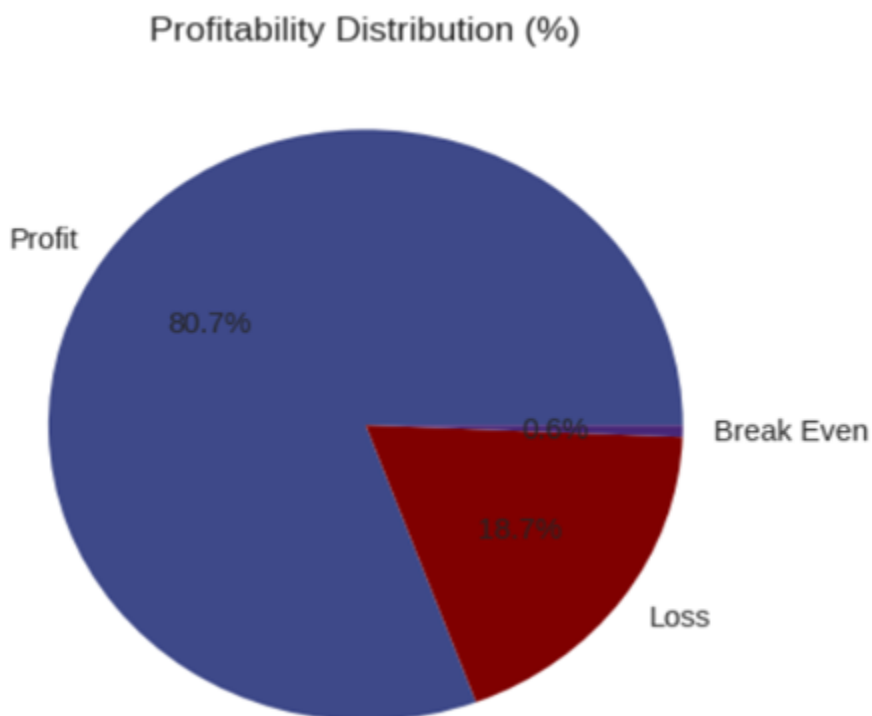
- Missing value analysis
- Duplicate removal
- Data type conversion
- Feature engineering
- Categorical variable encoding

Additional features such as delivery delay, order processing time, profitability indicators, and time-based variables were created to enhance analysis .

Exploratory Data Analysis

Profitability Analysis

The profitability distribution analysis revealed that most orders generated positive profit, while a smaller percentage resulted in losses.



Key Insight

The business operates profitably overall; however, specific orders and operational segments contribute disproportionately to losses.

Delivery Delay Analysis

Delivery performance was analyzed to understand the distribution and frequency of delayed shipments.



Key Insight

A significant portion of delays were concentrated within specific delay ranges, indicating recurring operational bottlenecks.

Profit Impact of Delays

The relationship between delivery delays and profitability was examined.

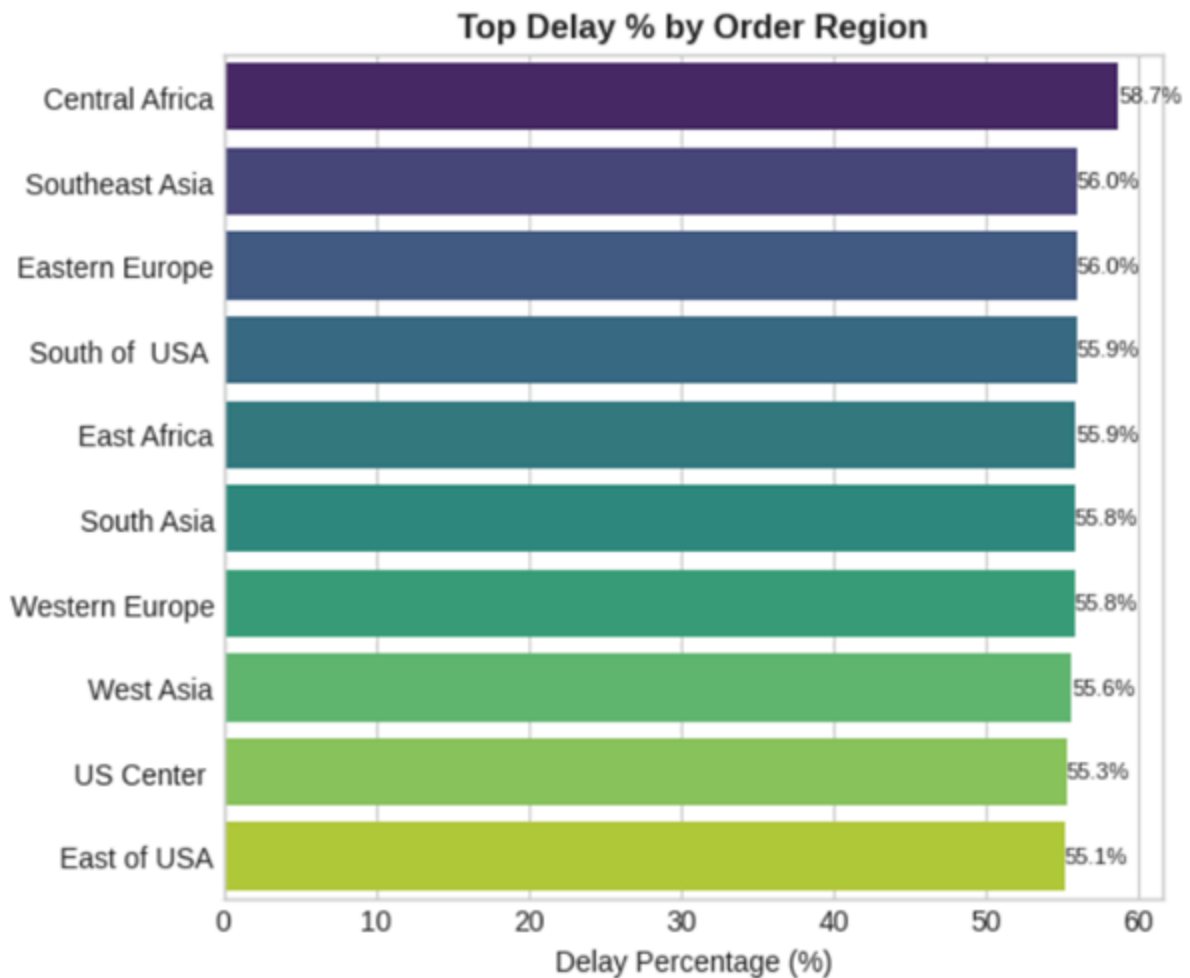


Key Insight

Longer delivery delays were associated with reduced average profitability, highlighting the financial impact of logistics inefficiencies.

Regional Performance Analysis

Regional delivery performance was evaluated to identify areas with higher delay rates.

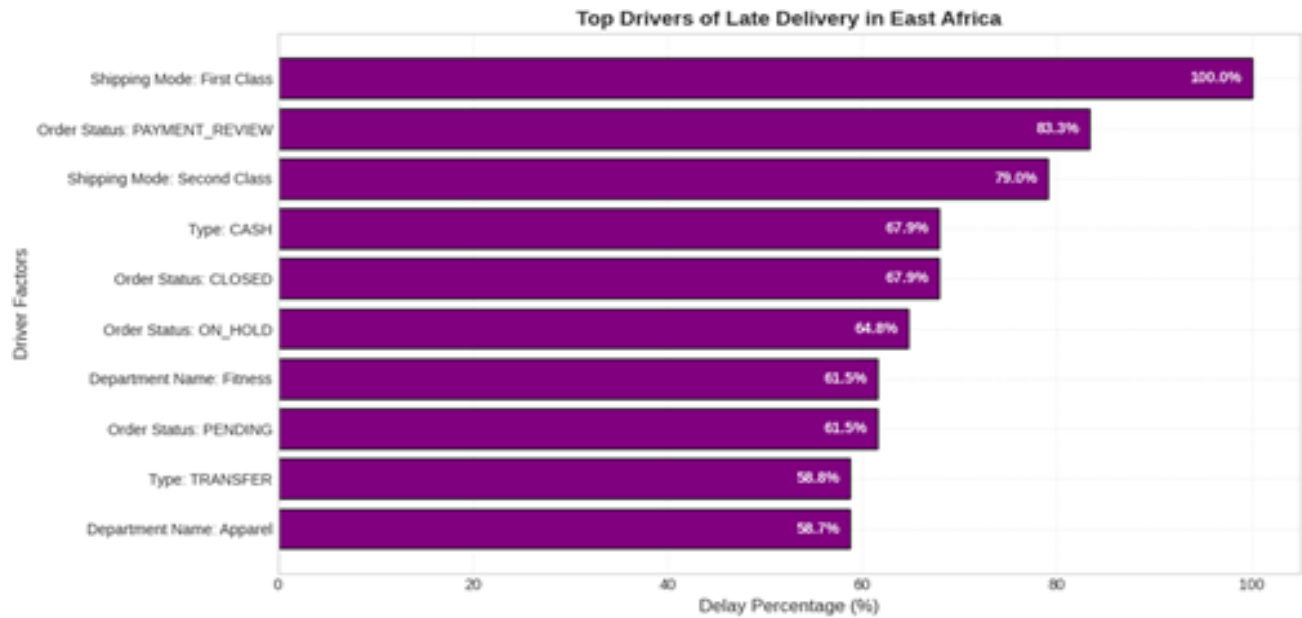


Key Insight

Certain regions consistently experienced higher late delivery percentages, suggesting opportunities for operational improvements and better logistics planning.

Delay Driver Analysis

The primary drivers of delayed deliveries were analyzed across multiple operational dimensions.



Major Drivers Identified

- Shipping Mode
- Customer Segment
- Department Name
- Order Status
- Order Type

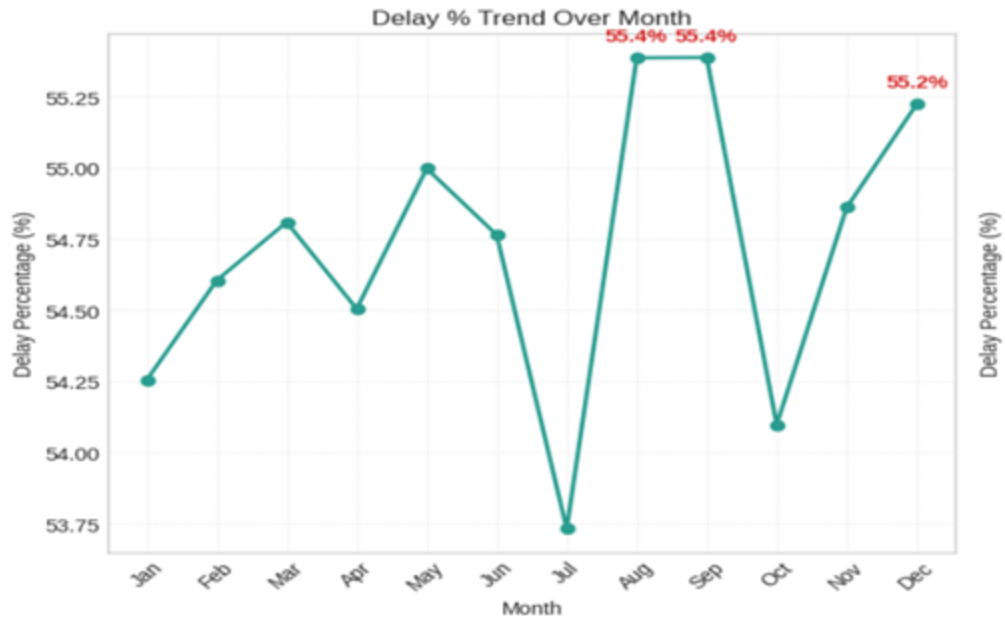
Key Insight

Shipping methods and operational processes were among the strongest contributors to delivery delays.

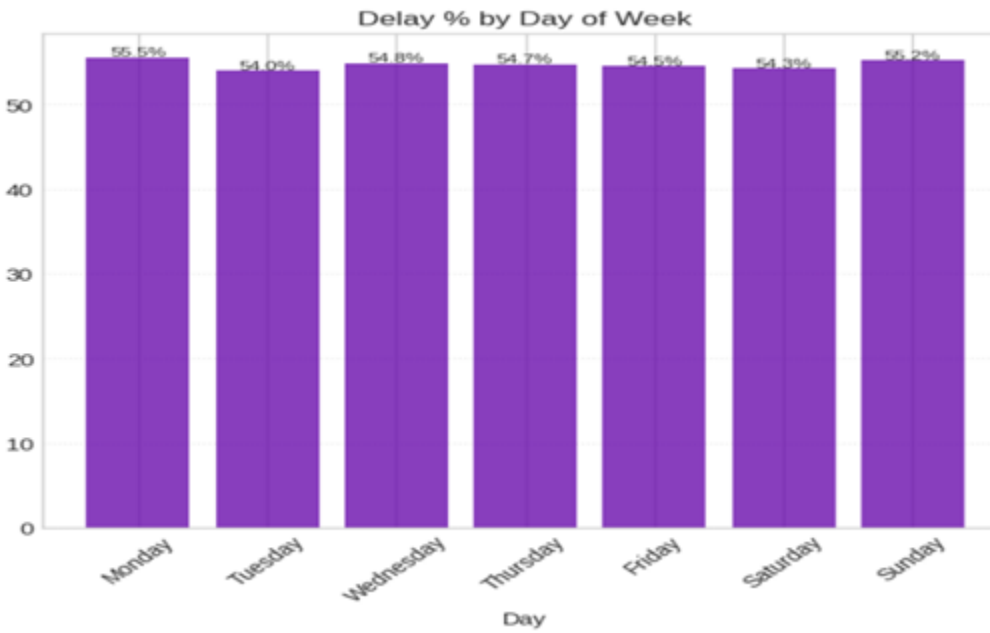
Time-Based Analysis

Delivery performance was analyzed across different time dimensions

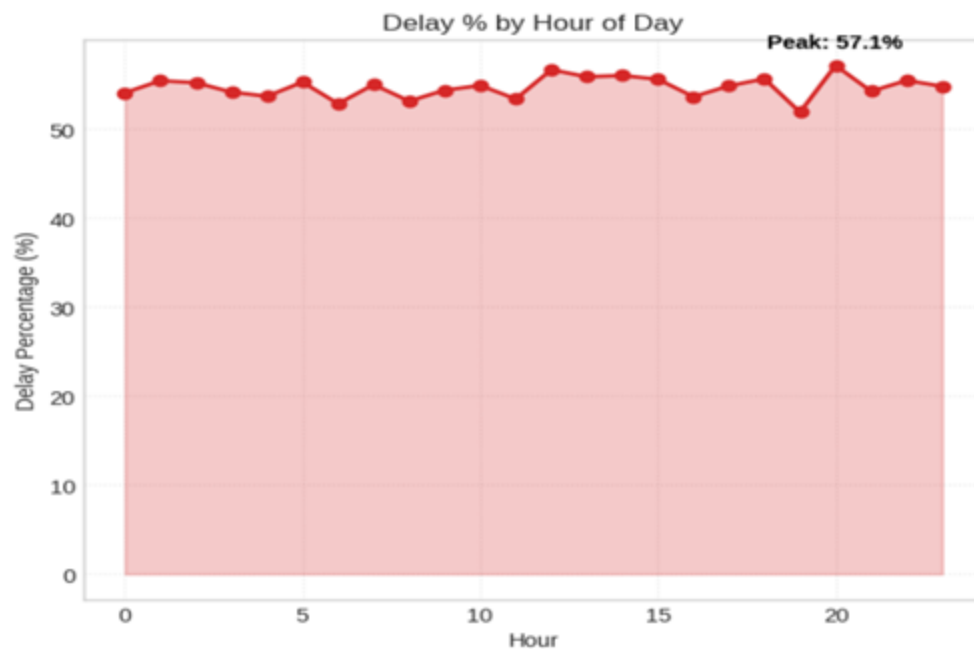
Monthly Analysis



Weekly Analysis



Hourly Analysis



Key Insight

Delay rates varied across months, weekdays, and order hours, indicating workload and operational capacity challenges during specific periods.

Machine Learning Model

Objective

Predict whether an order is likely to be delivered late.

Model Used

Random Forest Classifier

Data Preparation

- Feature Encoding
- Train-Test Split
- SMOTE Balancing

Evaluation Metrics

Accuracy	0.7247
Precision	0.8932
Recall	0.59
F1 Score	0.7106

Key Insight

The model successfully identified patterns associated with delayed deliveries and can support proactive logistics management.

Key Findings

1. Delivery delays significantly impact business profitability.
2. Certain shipping modes are associated with higher delay rates.
3. Regional performance varies considerably across the supply chain network.
4. Time-based patterns reveal operational bottlenecks.
5. Predictive analytics can help identify high-risk deliveries before delays occur.

Recommendations

1. Optimize High-Risk Shipping Methods

Improve transportation planning for shipping modes associated with frequent delays.

2. Focus on Underperforming Regions

Implement region-specific improvement strategies.

3. Deploy Predictive Monitoring

Use machine learning predictions to identify and manage high-risk orders.

4. Improve Operational Planning

Allocate resources effectively during peak demand periods.

5. Monitor Loss-Making Orders

Investigate operational factors contributing to profitability losses.

Conclusion

This project demonstrates how data analytics and machine learning can improve supply chain performance by identifying operational inefficiencies, reducing delivery delays, and supporting data-driven decision-making.